INCH-POUND
MIL-PRF-32187A
29 August 2017
SUPERSEDING
MIL-PRF-32187
24 February 2005

PERFORMANCE SPECIFICATION

GASKET, HYBRID, FLAT AND RAISED FACE FLANGE, FIRE RESISTANT, SHIPBOARD USE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers hybrid gaskets designed for sealing ASME B16.5, class 150, flat and raised face flanges in fuel and lube oil piping systems. Hybrid gaskets are fire resistant and are intended for shipboard use. Hybrid gaskets are defined as gaskets fabricated using two or more materials, usually an elastomeric compound serving as the primary seal and an enveloping graphite material serving as the fire-resistant seal.

1.2 Classification. Hybrid gaskets are of the following types and sizes, as specified (see 6.2).

1.2.1 Types. The types of hybrid gaskets are as follows:

a. Type I - Two circular, concentric seals recessed into the surface of a metal ring; an inner elastomeric seal (primary seal) and an outer flat graphite sheet material seal (fire-resistant seal).

b. Type II - Two circular, concentric seals recessed into the surface of a metal ring; an inner elastomeric seal (primary seal) and an outer spiral wound graphite seal (fire-resistant seal).

c. Type III - Multiple circular, concentric seals of sheet graphite material, recessed into the surface of a self-locating metal ring.

1.2.2 Sizes. Hybrid gasket sizes accommodate piping flanges with the following nominal pipe sizes (NPSs) (in inches): ¼, ½, ¾, 1, 1¼, 1½, 2, 2½, 3, 3½, 4, 5, 6, 8, 10, and 12.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5330

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

1.3 Part or identifying number (PIN). PINs to be used for hybrid gaskets acquired to this specification are created as follows:

M	**32187**	**-**	**X**	**X**
Prefix for Military Specification	Specification Number		Type (see code below)	Size (see code below)

Type	PIN Code
I	1
II	2
III	3

Size (Inches)	PIN Code
¼	A
½	B
¾	C
1	D

Size (Inches)	PIN Code
1¼	E
1½	F
2	G
2½	H

Size (Inches)	PIN Code
3	J
3½	K
4	L
5	M

Size (Inches)	PIN Code
6	N
8	P
10	Q
12	R

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN PETROLEUM INSTITUTE (API)

API STD 607 - Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats

(Copies of this document are available online at www.api.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 Metric/Inch Standard

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A106/A106M	-	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM D512	-	Standard Test Methods for Chloride Ion in Water
ASTM D1179	-	Standard Test Methods for Fluoride Ion in Water
ASTM D1246	-	Standard Test Method for Bromide Ion in Water
ASTM D3223	-	Standard Test Method for Total Mercury in Water
ASTM D3557	-	Standard Test Methods for Cadmium in Water
ASTM D3559	-	Standard Test Methods for Lead in Water
ASTM D4190	-	Standard Test Method for Elements in Water by Direct-Current Plasma Atomic Emission Spectroscopy
ASTM D4327	-	Standard Test Method for Anions in Water by Suppressed Ion Chromatography

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Hybrid gaskets furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Hybrid gaskets for both flat and raised face flanges shall be qualified.

3.2 Materials. Materials shall exhibit fluid compatibility satisfactory for long-term exposure to hydrocarbons such as fuel oils and lube oils. Materials shall not break down or break apart contributing particles into system fluids. Materials shall exhibit galvanic compatibility when installed in contact or close physical proximity to carbon steel, corrosion-resisting steel, bronze, or copper-nickel alloys.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

3.2.1 Detrimental materials. Detrimental materials shall not exceed the maximum concentrations, as specified in table I (see 4.5.1.3).

TABLE I. Detrimental materials.

Material	Maximum concentration permitted (parts per million [ppm])
Asbestos	None permitted
Halogens (total of chloride, fluorine, and bromide) 1/	1,000
Lead	None permitted
Cadmium plating	None permitted
Mercury and compounds (none intentionally added) 2/	10
Sulfur	10,000

NOTES:

1/ Materials containing more than 1,000 ppm total halogens are acceptable, provided only fluorine exceeds 250 ppm and total leachable halides do not exceed 250 ppm.

2/ During manufacture, fabrication, handling, packaging, and packing, hybrid gasket materials shall not come in contact with mercury or mercury compounds.

3.2.2 Toxicity and prohibited materials.

3.2.2.1 Toxicity. When evaluated in accordance with 4.6, the hybrid gasket shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.6 and 6.7).

3.2.2.2 Prohibited materials. The hybrid gasket shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. Hybrid gasket design and construction may vary from types listed in 1.2.1, provided that the requirements of this document are met. Unless otherwise specified (see 6.2), hybrid gasket sizes shall be in accordance with ASME B16.5, class 150 flanges with NPSs listed in 1.2.2.

3.3.1 Service life. Hybrid gaskets shall be designed for a minimum service life of 15 years.

3.4 Physical requirements.

3.4.1 Fire resistance. Hybrid gaskets shall be fire resistant to withstand a 30-minute exposure to a modified API STD 607 fire test (see 4.5.1.1).

3.4.2 Structural characteristics. Hybrid gaskets shall maintain system integrity when installed in flanges exhibiting customary bolt torques, flange face finishes, and allowable degrees of misalignment. When installed, hybrid gasket compression shall not damage the surface finish of the mating flange faces.

3.4.2.1 Gasket thickness. Hybrid gaskets shall have a maximum initial uncompressed thickness of 0.18 inch. Gasket thickness shall not exceed 0.135 inch under the compressive load associated with ASME B16.5 flange configuration and torque values applied to the size and number of corresponding bolting.

3.4.3 Sealing characteristics. Hybrid gaskets shall show no visible leakage during normal system operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

3.5 Performance requirements.

3.5.1 Operating environment. Hybrid gaskets shall prevent flange leakage in shipboard fuel and lube oil systems (excluding hydraulic systems) with operating temperatures of -20 to 250 degrees Fahrenheit (°F). Hybrid gasket sealing integrity shall not be affected by thermal cycling (see 4.5.1.2).

3.6 Cleanliness requirements. The surface of gaskets and related parts shall be commercially clean; for example, free of dirt, oils, grease, chips, and foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

Table II shows the relationship between the requirements in section 3 and the verifications in section 4 required for qualification and conformance inspection.

TABLE II. Examinations and inspections.

Test requirement	Requirement paragraph	Verification paragraph	Qualification	Conformance
Examination	3.2 through 3.6	4.4	X	X
Fire test	3.4.1	4.5.1.1	X	
Pressure and temperature exposure test	3.5.1	4.5.1.2	X	
Detrimental materials	3.2.1	4.5.1.3	X	
Toxicity and prohibited materials	3.2.2	4.6	X	X

4.2 Qualification inspection. Qualification inspection shall be performed on hybrid gaskets and their materials as specified herein. This inspection shall include the examination of 4.4, the tests of 4.5.1.1 through 4.5.1.3, and the assessment of 4.6, as specified in table II.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the assessment of 4.6, as specified in table II.

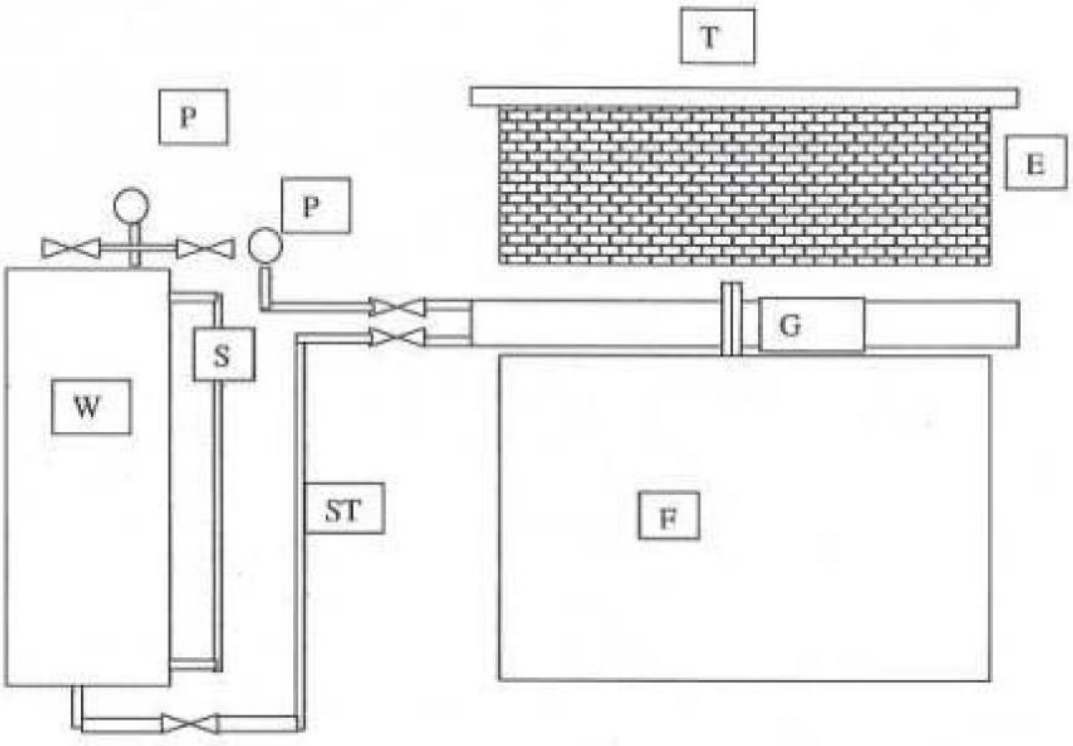
4.4 Examination. Each hybrid gasket shall be examined for compliance with the requirements specified in 3.2 through 3.6. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum performance shall constitute cause for rejection.

4.5 Methods of inspection.

4.5.1 Product certification. Hybrid gaskets of each size (1-inch, 4-inch, and 12-inch) shall be tested in both flat faced flanges and raised face flanges of the corresponding sizes. Gasket material samples shall be tested as applicable to 4.5.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

4.5.1.1 Fire test. The hybrid gasket shall be tested in accordance with and shall meet the maximum allowable external leakage rate acceptance criteria of API STD 607, as modified below. The test encompasses pressurizing the specimen with water and exposing it to a hydrocarbon temperature curve as specified in the standard. During the test, leakage rate is measured by monitoring source tank levels. Testing of each size hybrid gasket shall be conducted using flat and raised face flanges and shall meet the following criteria. Figure 1 illustrates a typical fire test setup. Any variance in setup or test procedures requires NAVSEA approval prior to testing.



F – Furnace

G – Gasket/Pipe/Flange Specimen

S – Sight Gauge

W – Water Source Tank

T – Furnace Top

E – Furnace Extension Block

P – Pressure Gauge with Steam Trap

ST – Steam Trap

⋈ – Valve

FIGURE 1. Typical modified API STD 607 fire test setup for hybrid gasket.

4.5.1.1.1 Testing criteria. This 30-minute fire exposure test is intended to demonstrate that the gaskets will not fail catastrophically or contribute a significant amount of combustible liquid to a fire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

4.5.1.1.2 Test sample. Six gaskets, two 1-inch, two 4-inch, and two 12-inch NPS, shall be tested, one sample of each size in flat face flanges and one sample of each size in raised face flanges. Each shall be installed in the applicable flange system including requirements for flange design and material, bolt design and material, torque limits in accordance with figure 1 and the following. Piping shall be seamless carbon steel (ASTM A106/A106M, grade A or B). Flanges shall be ASME B16.5, class 150, socket weld, carbon steel with faces having a phonographic surface finish of 250 micro-inches Roughness Average (Ra). Flange bolts shall be MIL-DTL-1222, grade 5, carbon steel, ½-inch diameter for 1-inch flanges, ⅝-inch diameter for 4-inch flanges, and ⅞-inch diameter for 12-inch flanges. Bolts shall be torqued to 94 foot-pounds (ft-lbs) for 1-inch NPS flanges, 187 ft-lbs for 4-inch flanges, and 582 ft-lbs for 12-inch flanges. To reduce furnace heat sink effects, all piping except 6 inches of pipe each side of the flange may be covered with 1-inch thick, 6-pounds per cubic foot ceramic fiber insulation.

4.5.1.1.3 Test conditions. The gasket/flange assemblies shall be subjected to the test conditions of API STD 607 as modified herein:

a. Substitute the test gasket/flange assembly for the test valve throughout API STD 607.

b. The test pressure shall be 100 pounds per square inch gauge (psig), instead of the 30 psig used in the Fire-Test Conditions described in API STD 607.

c. Delete through-valve leakage collection of API STD 607.

d. As a further clarification, the Test Procedure in API STD 607 applies to the flange/gasket assembly. External leakage is calculated for the total fire test period. The total fire test period is the fire exposure test period (30 minutes) plus the time to cool down the assembly to below 212 ºF.

e. Delete valve operational test of API STD 607.

f. Delete through-valve seat leakage of API STD 607.

4.5.1.1.4 Acceptance criteria. The flange/gasket assembly shall meet the maximum allowable external leakage rate acceptance criteria of API STD 607 for the total fire test period. External leakage for flange/gasket assembly NPS 10 or smaller for the total fire test period, including cool down, shall not exceed 1.5 cubic inches (25 milliliters) per minute per NPS size. For flange/gasket assembly larger than NPS 10, the maximum allowable external leakage shall be 15 cubic inches (250 milliliters) per minute (regardless of flange/gasket size). The total fire test period is the duration of the 30-minute fire test plus the cool-down period to reach 212 ºF. The through-valve seat leakage of API STD 607 is not required.

4.5.1.1.5 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the NAVSEA Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) or International Code Council's International Accreditation Services (IAS). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.6).

4.5.1.2 Pressure and temperature exposure test. The hybrid gasket shall meet the following pressure and temperature exposure test requirements.

4.5.1.2.1 Test sample. Six gaskets, two 1-inch, two 4-inch, and two 12-inch NPS, shall be tested, one sample of each size in flat face flanges and one sample of each size in raised face flanges. Each shall be installed in the applicable flange system including requirements for flange design and material, bolt design and material, and torque limits in accordance with the following. Piping shall be seamless carbon steel (ASTM A106/A106M, grade A or B). Flanges shall be ASME B16.5, class 150, socket weld, carbon steel with faces having a phonographic surface finish of 250 micro-inches Ra. Flange bolts shall be MIL-DTL-1222, grade 5, carbon steel, ½-inch diameter for 1-inch flanges, ⅝-inch diameter for 4-inch flanges, and ⅞-inch diameter for 12-inch flanges. Bolts shall be torqued to 94 ft-lbs for 1-inch NPS flanges, 187 ft-lbs for 4-inch flanges, and 582 ft-lbs for 12-inch flanges.

4.5.1.2.2 Test conditions. The gasket/flange assemblies shall be subjected to the test conditions as follows:

a. Clean flange face and install test hybrid gasket in test assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

b. Where a pump/regulator is not used to achieve and maintain pressure, calculate the amount of lube oil required to fill the flange assembly when the test conditions are at least 150 psig. Place a measured amount of lube oil, slightly greater than required in the flange assembly prior to closing.

c. Torque the flange assembly using the proper bolt tightening sequence.

d. Slowly heat the flange assembly to at least 250 ºF but not to exceed 270 ºF monitoring flange temperature. Since the flange may heat up faster than the lube oil, flange temperature shall be carefully monitored to ensure overheating does not occur.

e. If necessary, add small quantities of lube oil to the test assembly until at least 150 psig is obtained.

f. Monitor for at least 4 hours, checking for leakage and pressure loss. Use an ultrasonic leak detector or fluorescent dye to assist with locating leakage.

g. If leakage occurs, re-torque the flange bolts. Adjust pressure to 150 psig and verify tight seal for a 4-hour period.

h. Reduce the temperature to ambient, repeat steps d through g, and then reduce the temperature to ambient again.

4.5.1.2.3 Acceptance criteria. The hybrid gasket shall show no visible leakage throughout the duration of the test and shall show no signs of leakage in the 30-minute period directly following the test. If the hybrid gasket leaks or fails to hold pressure after two re-torques, the gasket shall be considered inadequate.

4.5.1.3 Detrimental materials.

4.5.1.3.1 Heavy metals. Heavy metals shall be analyzed in accordance with the methods specified in ASTM D3223, D3557, D3559, or D4190 (see 3.2.1).

4.5.1.3.2 Water leachable halides. For determination of water leachable halides, the concentration of bromide, chloride, and fluoride ions in each filtrate shall be determined in accordance with ASTM D1246, D512, D1179, and D4327, as appropriate, and shall be corrected by the results of a blank determination (see 3.2.1).

4.6 Toxicity and prohibited materials. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.2.1 and 3.2.2.2, as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the hybrid gaskets using data provided by the manufacturer/distributor to the NMCPHC (see 3.2.2.1 and 6.7).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hybrid gaskets are intended for use in sealing flat and raised face flanges in fuel and lube oil piping systems installed on Navy surface ships and submarines. Hybrid gaskets are not approved for services requiring Level I and SUBSAFE levels of essentiality. Existing stock of hybrid gaskets manufactured to MIL-PRF-32187 dated 24 February 2005 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size of hybrid gasket (see 1.2).

c. Quantity required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

d. Applicable National Stock Number (NSN).

e. Specific issues of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).

f. If the hybrid gasket size is other than as specified (see 3.3).

g. Additional fire testing provisions (see 4.5.1.1.5 and 6.6).

h. Packaging requirements (see 5.1).

i. Shelf-life requirements (see 6.5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32187 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Flange specification. The following documents provide useful information regarding guidance for bolting, flange geometry, flange materials, and application:

MIL-PRF-20042	-	Flanges, Pipe, Bronze (Silver Brazing)
MIL-STD-438	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships

6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil.

6.6 Additional fire testing provisions. NAVSEA reserves the right to witness the tests, and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. NAVSEA will provide a letter upon request that indicates the approved laboratories which are accredited to ISO/IEC 17025 and which details how to obtain and maintain accreditation (see 6.2).

6.7 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.8 Subject term (key word) listing.

Flat Face Flange

Fuel Oil Piping

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.

Hybrid Gasket

Lube Oil Piping

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 5330-2017-003)

Review activities:
Army – AV, MR
Navy – CG, SA
DLA – IS
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:10Z
Check the source to verify that this is the current version before use.